

1.1 Probability and Uncertainty

*my philosophical understanding
(not important for applications)



(a) Data is random

Probability is long-run frequency, fraction of an outcome in sequence of repetitions, e.g. dice roll

Frequentists: p-value, χ^2 test, confidence intervals...

(b) Epistemic

Probability is state of your knowledge, degree of uncertainty (probability of rain tomorrow 60%).

Bayesian statistic:

- naturally incorporates both (a) and (b)
- full uncertainty quantification
- sequential, makes use of prior knowledge
- intuitive, matches with how we reason
- scales to many parameters

1.2

Bayes theorem

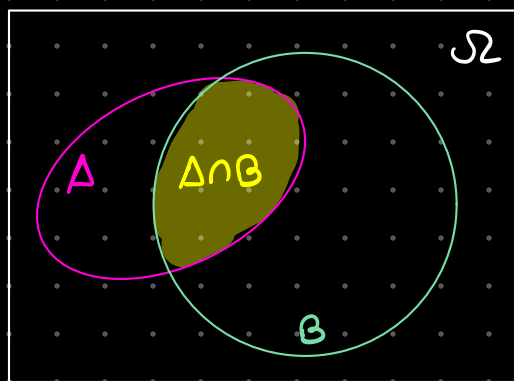
More formally:

For a fixed set Ω of all outcomes the **probability** of an event (subset of Ω) is function $P(A) : \text{events} \rightarrow [0, 1]$ satisfying 3 axioms: $P(A) \geq 0 \forall A$, $P(\Omega) = 1$, $P(A_1 \cup A_2 \dots) = P(A_1) + P(A_2) \dots$

Conditional probability $P(A|B)$ - probability of A occurring if B is assumed to have happened

$$P(A|B) := \frac{P(A \cap B)}{P(B)}$$

($P(A \cap B)$ renormalize to $P(B)$)



$$P(B \cap A) = P(B|A) P(A)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \overset{\substack{\text{same} \\ \text{set}}}{=} \frac{P(B \cap A)}{P(B)} \overset{\downarrow}{=} \frac{P(A) P(B|A)}{P(B)}$$

Say $A = \text{model}$, $B = \text{observed data} \Rightarrow$

$$\underbrace{P(\text{model} | \text{data})}_{\text{posterior}} = \frac{\underbrace{P(\text{model})}_{\text{prior}} \underbrace{P(\text{data} | \text{model})}_{\text{likelihood}}}{\underbrace{P(\text{data})}_{\text{evidence}}}$$

Posterior - conditional probability $P(\text{model} | \text{data})$
 probability of a model given data

- "How much credence should I give to a model in the light of the new data?"

Prior - assumed probability (distribution) of a model before the data.

- "How much did I believe it beforehand?"

Likelihood - conditional probability $P(\text{data} | \text{model})$
 probability of seeing the data given a model

- "If the model is true, how probable is the data we got?"

Marginal likelihood - evidence - likelihood that has been marginalized (integrated) over the models (model parameter space).

Normalization factor in Bayes theorem

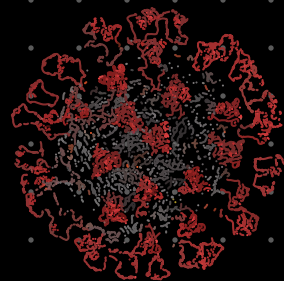
- „How probable is the data under all the models (parameters) together?“

An update process!

To visualize discrete case we can use Bayes table

- see `01_bayesian_thinking.ipynb`

Example 1.1 - Medical test



A rare disease affects 1% of population. Test gives false positive only in 5% cases and never misses a real case.

Patient tests positive, what is probability it's real?

$$\text{Prior: } P(\text{sick}) = 0.01 \Rightarrow P(\text{healthy}) = 0.99$$

$$\text{Likelihoods: } \begin{cases} P(\text{positive} | \text{healthy}) = 0.05 \\ P(\text{positive} | \text{sick}) = 1 \end{cases}$$

$$\text{Evidence: } p(\text{positive}) = P(\text{sick}) P(\text{positive} | \text{sick}) + P(\text{healthy}) P(\text{positive} | \text{healthy})$$

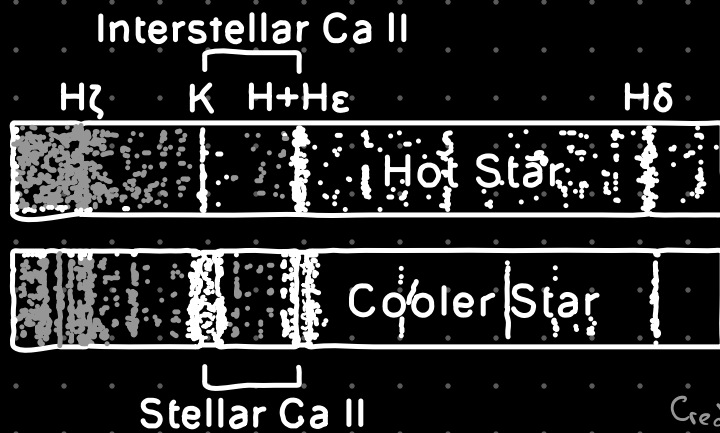
$$P(\text{sick} | \text{positive}) = \frac{P(\text{sick}) P(\text{positive} | \text{sick})}{P(\text{positive})} = 0.162$$

$$P(\text{healthy} | \text{positive}) = \frac{P(\text{healthy}) P(\text{positive} | \text{healthy})}{P(\text{positive})} = 0.832$$

$\Rightarrow$ only 16%

Examples 1.2 and 1.3
are in `op_bayesian_thinking.ipynb`

Note on 1.3.6)



Credit: Cohen M.L.

$$\begin{aligned} P(\text{Ca II} \mid H \mid \lambda, B-V) &\propto P(\text{Ca II} \mid H) P(\lambda, B-V \mid \text{Ca II} \mid H) = \\ &= \underbrace{P(\text{Ca II} \mid H) P(\lambda \mid \text{Ca II} \mid H)}_{\text{new prior (not flat)}} P(B-V \mid \text{Ca II} \mid H) = \\ &= P^*(\text{Ca II} \mid H) P(B-V \mid \text{Ca II} \mid H) \end{aligned}$$

→ sequential update process!

Summary

Inference is an update process:

Belief after = Belief before $\times$ how well model explains the data,
divided by how well everything else explains it.

② Building simple inference model

$$\underbrace{P(\theta | \text{data})}_{\text{posterior}} = \frac{\underbrace{P(\theta)}_{\text{prior}} \underbrace{P(\text{data} | \theta)}_{\text{likelihood}}}{\underbrace{\int P(\theta') P(\text{data} | \theta') d\theta'}_{\text{evidence}}}$$

Now $P(\theta)$ and $P(\theta | \text{data})$ are probability densities

Example 2.1 - Chocolate buns

How much chocolate in your favourite buns?

You weigh 25 chocolate buns.

The pack promises you 20g per bun $\Rightarrow$



prior is $P(\mu) = \mathcal{N}(20, 2^2)$

amount of chocolate in the bun $\nearrow$ normal distribution with mean $\mu_0 = 20\text{g}$ and standard deviation $\sigma_0 = 2\text{g}$

Your scale determines likelihood function:

$$P(x_i | \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

measurement $\nearrow$ model parameters

σ is given in the scale manual $\sigma = 2.5\text{g}$

We want posterior: $P(\mu | \{x_i\})$

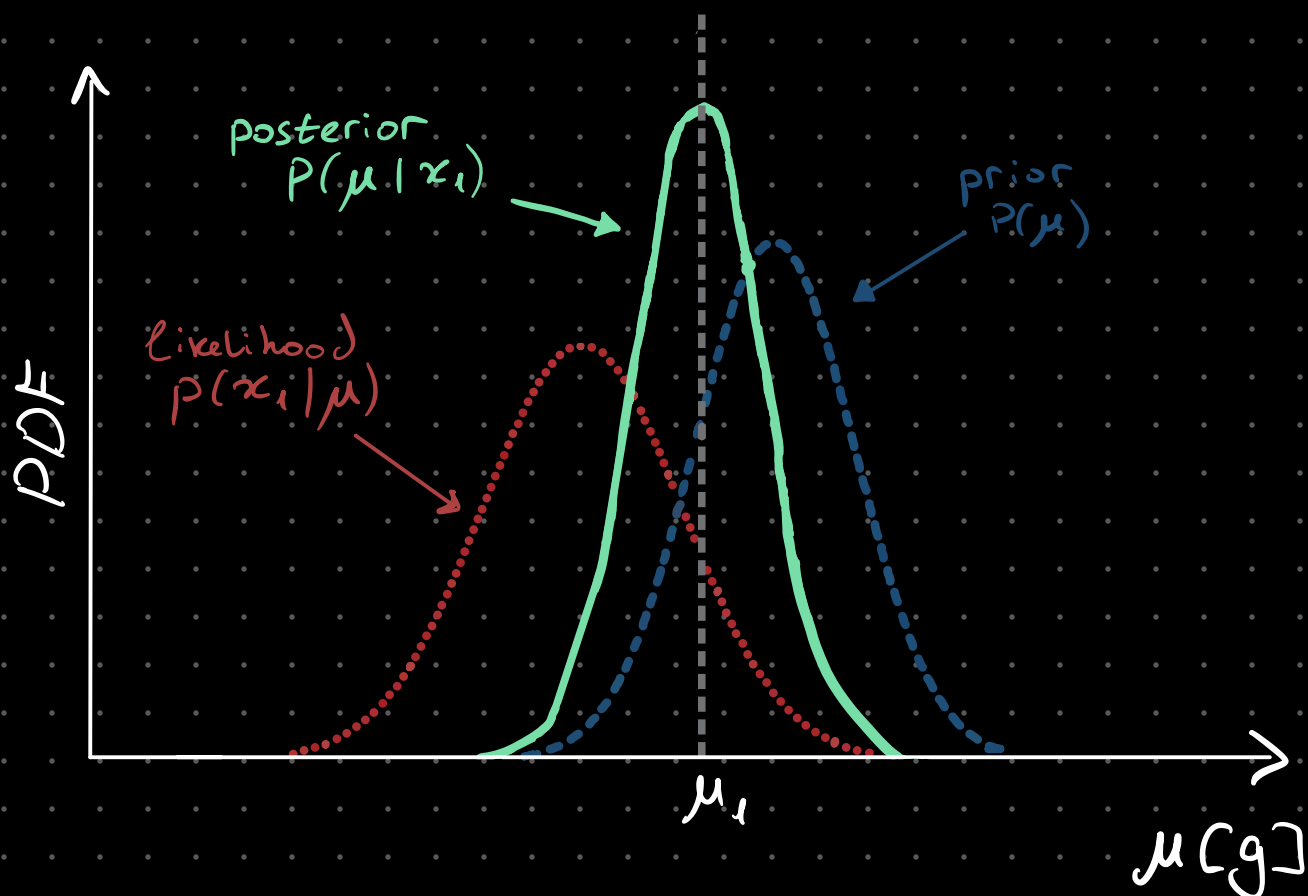
First weight we can do analytically: $x_1 = 15.2 \text{ g}$

$$P(\mu | x_1) = \frac{P(\mu) \cdot P(x_1 | \mu)}{\int d\mu' P(\mu') P(x_1 | \mu')} = \frac{\exp\left(-\frac{(\mu - \mu_1)^2}{2\sigma_1^2}\right)}{\int d\mu' \exp\left(-\frac{(\mu' - \mu_1)^2}{2\sigma_1^2}\right)} =$$

$$= \frac{1}{\sqrt{2\pi}\sigma_1} \exp\left(-\frac{(\mu - \mu_1)^2}{2\sigma_1^2}\right)$$

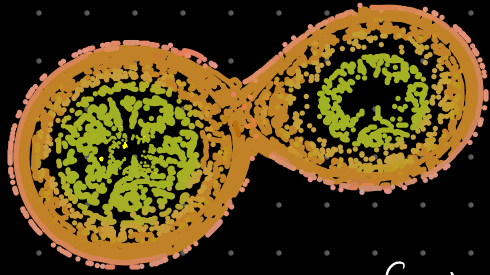
$$\text{where } \mu_1 = \sigma_1^2 \left(\frac{\mu_0}{\sigma_0^2} + \frac{x_1}{\sigma^2} \right) \quad \text{and} \quad \sigma_1^2 = \frac{\sigma_0^2 \sigma^2}{\sigma_0^2 + \sigma^2}$$

$$\Rightarrow \sigma_1^2 = 2.44 \text{ g}^2 \quad \mu_1 = 18.13 \text{ g}$$



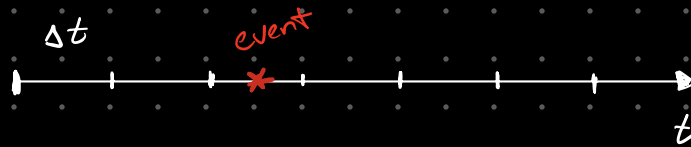
The rest of the problem in [02_simple_inference_models.ipynb](#)

Note on example 2.2



Poisson distribution

Credit: Rezzolla



$$P(\text{event}) = R \Delta t$$

$$P(\text{no event}) = 1 - R \Delta t$$

$$P(2 \text{ events}) = 0 \quad \text{for small } \Delta t$$

$\Rightarrow$ Binomial distribution - probability of k successes in n amount of trials

$$P(N=k) = C_n^k p^k (1-p)^{n-k}$$

$$p = R \Delta t = R \frac{1}{n} = \frac{\lambda}{n} \quad \leftarrow \text{expected \# of successes}$$

$$\Delta t \rightarrow 0 \Rightarrow n \rightarrow \infty \text{ as } \lambda \text{ - fixed}$$

$$P(k) = \frac{n!}{k!(n-k)!} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k}$$

$$\underbrace{\frac{n(n-1)\dots(n-k+1)}{n^k}}_{\rightarrow 1} \cdot \frac{\lambda^k}{k!} \cdot \underbrace{\left(1 - \frac{\lambda}{n}\right)^n}_{e^{-\lambda}} \cdot \underbrace{\left(1 - \frac{\lambda}{n}\right)^{-k}}_{\rightarrow 1} = \frac{\lambda^k}{k!} e^{-\lambda}$$

Summary

- Every summary number - mean CIs, universal relation - is read off the posterior distribution.
- An informative likelihood outweighs a reasonable prior, but the support of the posterior is fully contained in the support of the prior.

③ More dimensions

Example 3.1

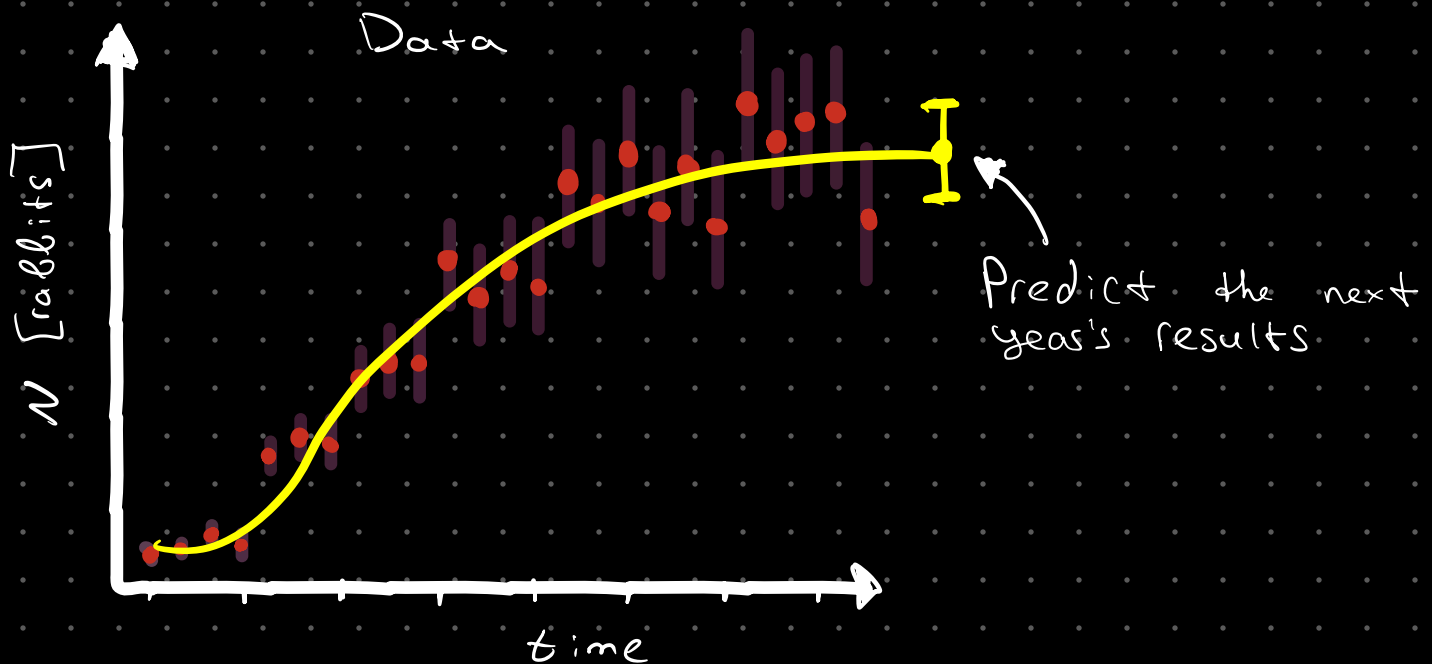


The rabbits population grows as

$$\frac{dN}{dt} = r N \left(1 - \frac{N}{K}\right)$$

↑ growth rate
↑ # of rabbits
← carrying capacity

⇒ exponential growth that saturates at K .



Survey every half a year determines likelihood:

$$p(r, K | \{N_i\}) \propto \underbrace{p(r) p(K)}_{\text{priors}} \exp \left[-\frac{1}{2} \sum_i \left(\frac{\overset{\text{data}}{N_i} - \overset{\text{model prediction}}{N(t_i; r, K)} }{\sigma_i} \right)^2 \right]$$

↑ parameters
↑ data

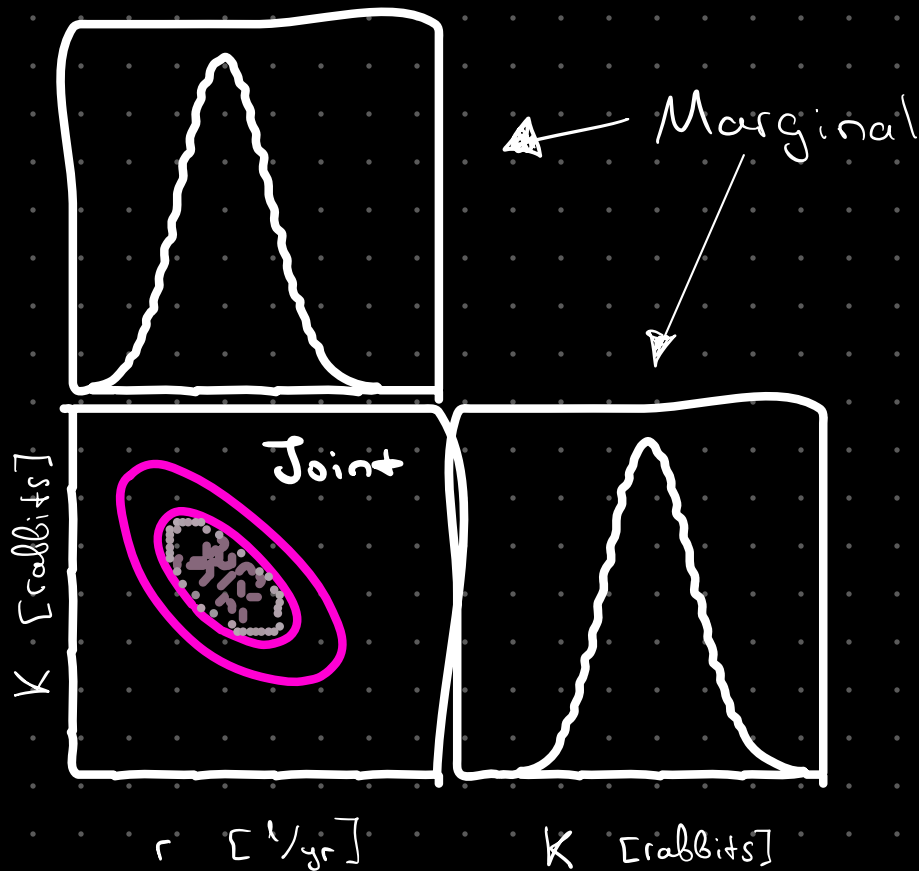
$p(r, K | \{N_i\})$ - joint posterior (includes correlation between parameters)

$$p(K | \{N_i\}) = \int dr p(r, K | \{N_i\})$$

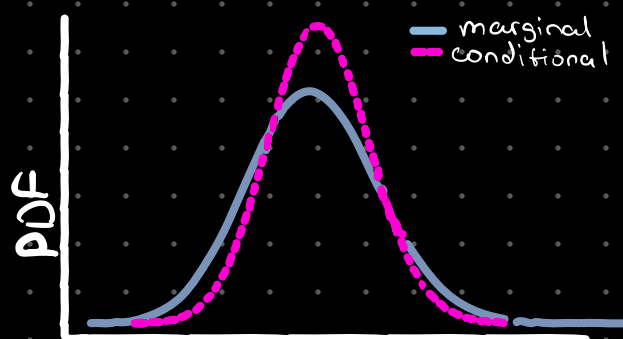
$$p(r | \{N_i\}) = \int dK p(r, K | \{N_i\})$$

Marginals

$p(K | r=r_0, \{N_i\})$ - conditional



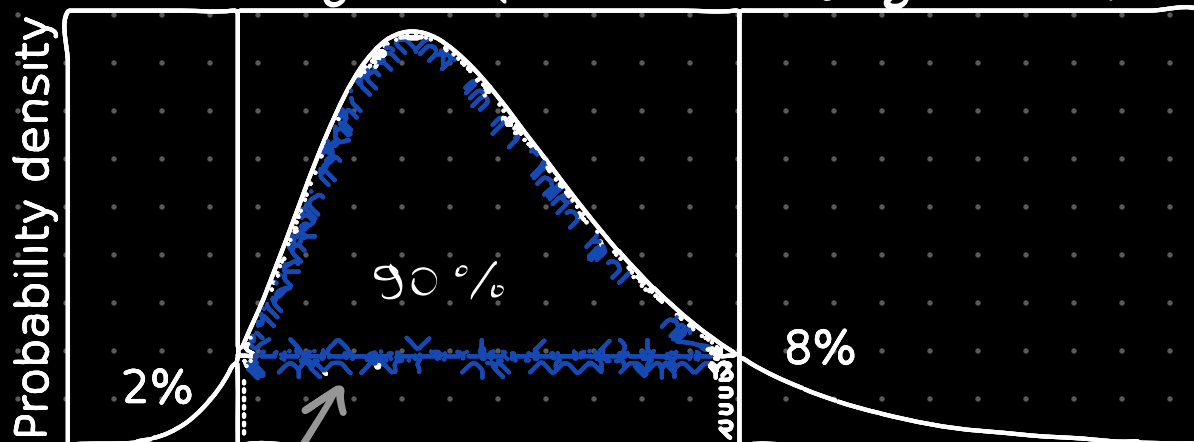
Nuisance parameters - unknown variable of the model that is not the primary focus of the study - can be integrated out (marginalized) propagating its uncertainty to the parameter of interest.



Credible intervals (CIs) $-\int_a^b P(\theta | \text{data}) d\theta = \#$

$\# = 0.68$ for 1σ and 0.95 for 2σ

highest posterior density (HPD)

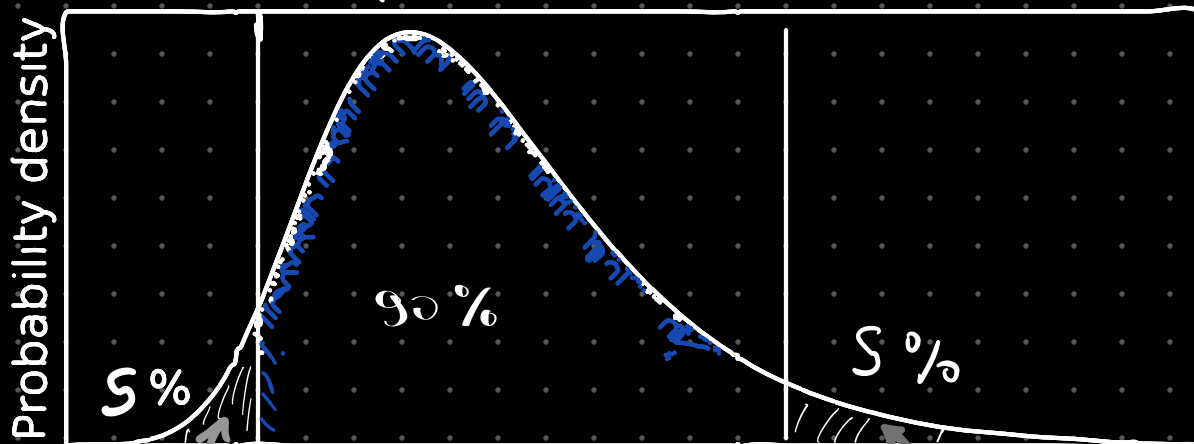


Value of the parameter

Credit: Wiki

Shortest interval holding 90%
(for multimodal might be disjoint)

equal-tailed (quantile)



Value of the parameter

equal probability in each tail
(only defined in 1D)

Quantiles and HPD are identical for
symmetric single-peaked distributions

Bayes factor

$$B = \frac{P(\text{Data} \mid \text{Model 1})}{P(\text{Data} \mid \text{Model 2})} = \frac{\int d\theta_1 \underbrace{P(\text{Data} \mid \theta_1)}_{\text{marginalized likelihood}} \underbrace{P(\theta_1 \mid \text{Model 1})}_{\text{prior on } \theta_1 \text{ with model 1}}}{\int d\theta_2 P(\text{Data} \mid \theta_2) P(\theta_2 \mid \text{Model 2})}$$

$$P(\text{Model 1} \mid \text{Data}) = \frac{P(\text{Data} \mid \text{Model 1}) P(\text{Model 1})}{\sum_{i=1}^2 P(\text{Data} \mid \text{Model } i) P(\text{Model } i)}$$

$$\Rightarrow B = \underbrace{\frac{P(\text{Model 1} \mid \text{Data})}{P(\text{Model 2} \mid \text{Data})}}_{\text{ratio of posteriors}} \cdot \underbrace{\frac{P(\text{Model 2})}{P(\text{Model 1})}}_{\text{inverse ratio of priors}}$$

$\Rightarrow$ prior odds $\cdot B =$ posterior odds

priors should be proper $\int d\theta_i P(\theta_i) = 1$

B is used to quantify which model the data prefer.

Jeffreys' scale:

B	
< 1	Model 2 is preferred
~ 1	Barely worth mentioning
~ 10	Substantial
~ 30	Strong
~ 100	Decisive

Summary

- In multi-parameter inference the result is the joint posterior distribution
- Marginalization is the way to propagate nuisance parameter's uncertainty into the answer
- The ratio of evidence, known as the Bayes factor, is used for model comparison

4 The Bayesian renaissance: MCMC

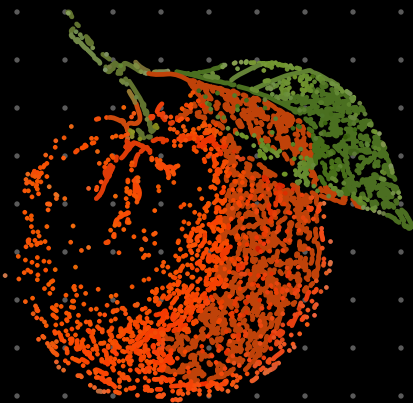
Realistic complex models $\Rightarrow$

- 1) posterior with no closed form
- 2) multi dimensional parameter space

For almost any application we need

multidimensional numerical integration $\Rightarrow$ Monte Carlo

$\rightarrow$ Can we get away with integration over grid or simple uniform sampling?



Assume posterior is spherical

(d-dimension orange)

What is the volume of the peel (5% width)?

$$V_d(R) = \frac{\pi^{d/2}}{\Gamma(d/2 + 1)} R^d$$

$$\Rightarrow V_{\text{peel}} = V_d(R) (1 - 0.95^d)$$

dimension	grid points (100 per axis)	volume of 5% peel
2	10^4	9.8 %
6	10^{12}	26.5 %
10	10^{20}	40.1 %
100	10^{200}	99.4 %

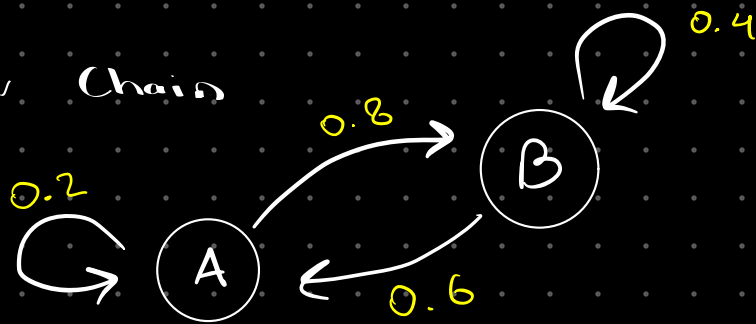


Not feasible at all



Need smart sampling

Markov Chain

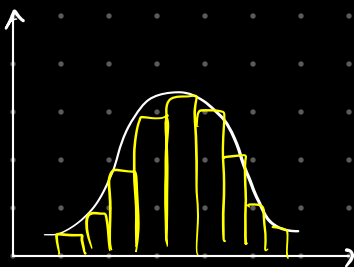


probability of each event depends only on previous event

+

Monte Carlo - repeated random sampling

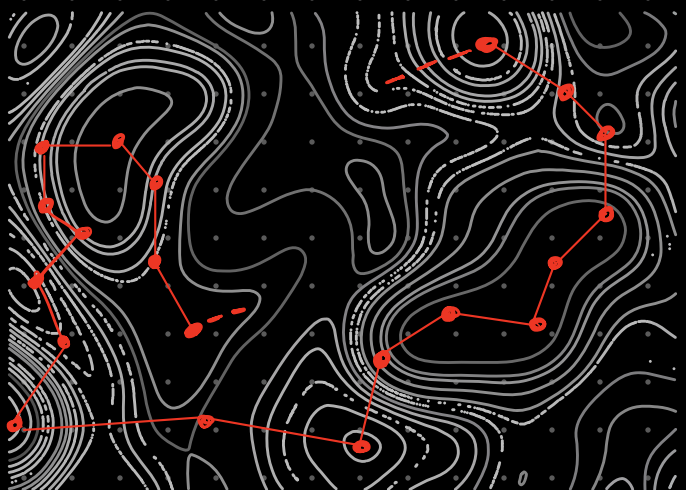
to approximate distribution.



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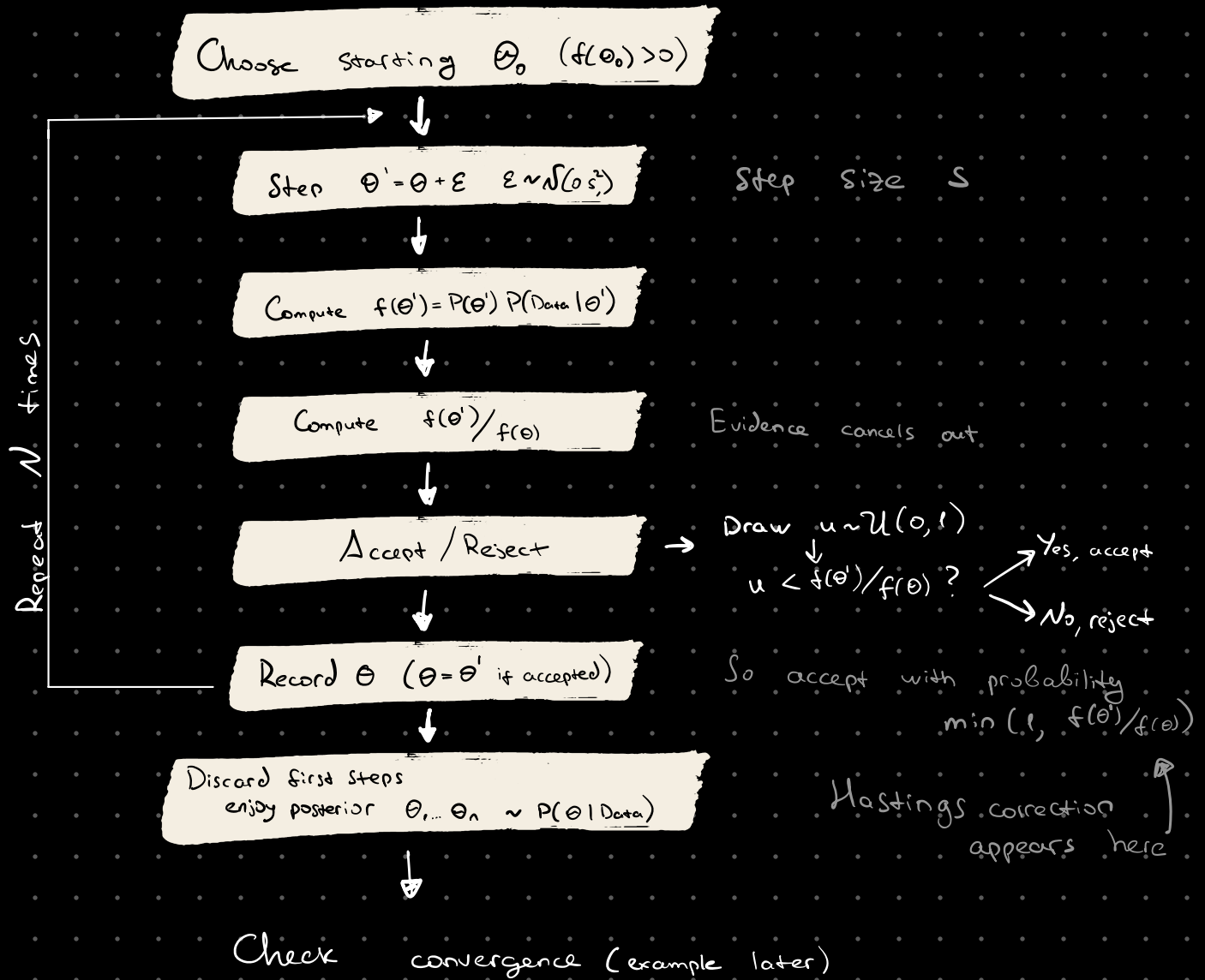
MCMC (in Bayesian context) - family of algorithms, whose long-run equilibrium approximates a posterior.

Random walk of a hiker in foggy mountains, who can feel the slope underfoot and is to spend more time at high altitude.



The classical example of such algorithm is **Metropolis - Hastings**

$$f(\theta) = P(\theta) P(\text{Data} | \theta) \propto \text{posterior}$$



Example 4.1 - foxes and rabbits

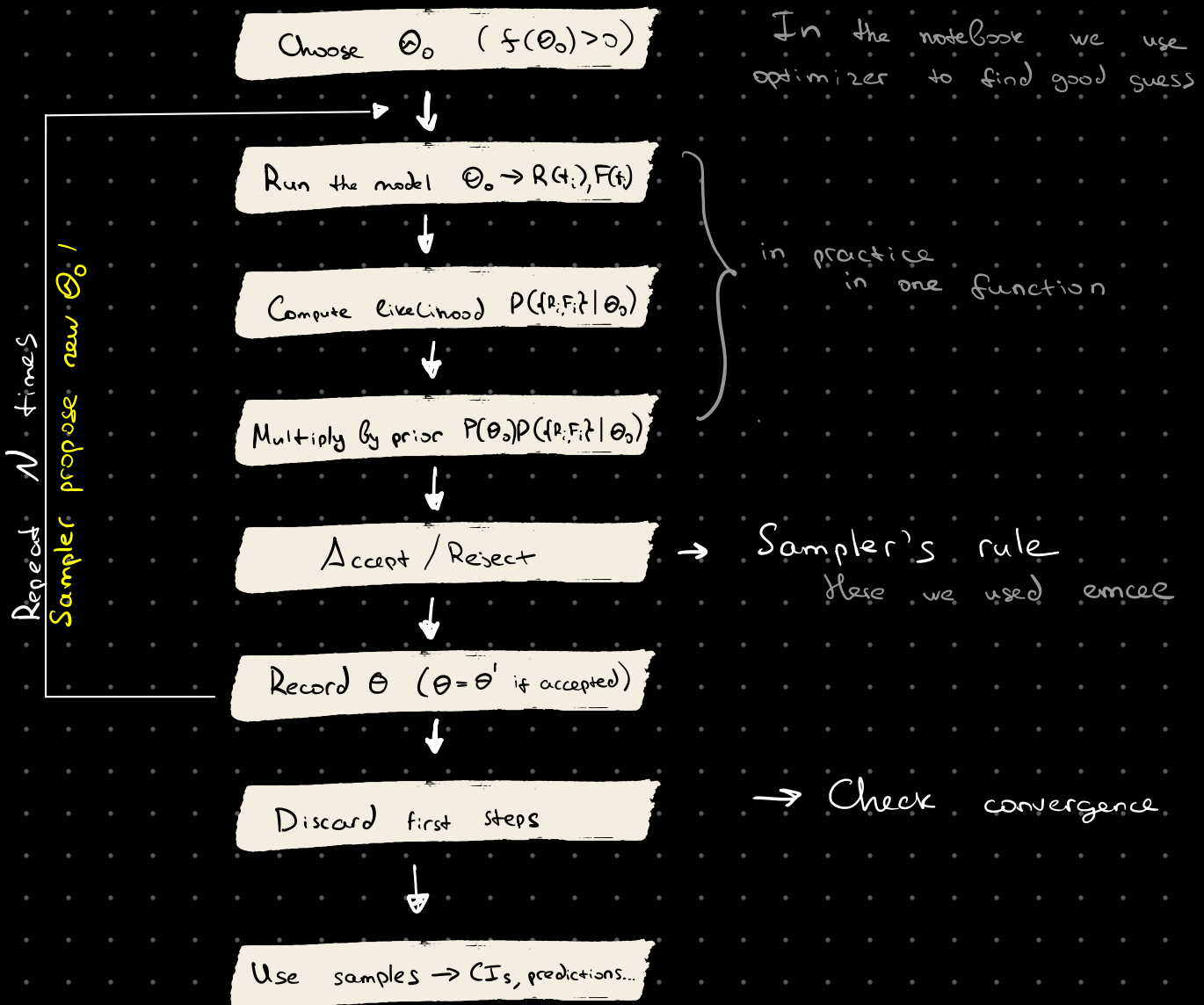
Full set-up in 04_mcmc.ipynb. Here is flowchart and summary.

Prior : flat box on $\theta = (\alpha, \beta, \delta, \gamma, R_0, F_0)$

Model : Lotka-Volterra - ODE $\theta \rightarrow R(t), F(t)$

Likelihood : $P(\{R_i, F_i\}|\theta) \propto \exp\left(-\frac{1}{2} \sum_i \left(\frac{R_i - R(t_i; \theta)}{\sigma_{R_i}}\right)^2 - \frac{1}{2} \sum_i \left(\frac{F_i - F(t_i; \theta)}{\sigma_{F_i}}\right)^2\right)$

Data : R_i, F_i i is one survey a year for 30 yr



Summary

- With MCMC we can do Bayesian inference on realistic, multidimensional problems with no closed forms.
- Samples of the posterior give everything we need
- Computation (better samplers, approximate inference, simulation-based inference) is one of the active areas of Bayesian research.

There is a summary of what I skipped in this course
in the 04-mcmc.ipynb



Thank you!

